

Equilibrium Displacement Models

Elasticities and Forecasting

Elasticities are useful tools in applied economics. They tell us:

- whether a tax on a specific good will raise large or small revenues,
- which goods are the greatest competitors, and
- how prices and quantities may change in response to shocks.

For example:

- How does the supply of pork decrease due to tougher regulations on manure treatment and disposal?
- How much will the demand for pork increase due to an increase in the price of beef?

We do not need to know the *exact* supply and demand curves. Estimates of elasticities are usually sufficient.

By incorporating elasticities into an **equilibrium displacement model (EDM)**, we can calculate price and quantity changes resulting from outside shocks.

Endogenous and Exogenous Variables

In an EDM, all variables are grouped as either **endogenous** or **exogenous**.

- **Endogenous variables:** Determined *within* the model (price and quantity).
- **Exogenous variables:** Determined *outside* the model and taken as given.
 - Examples: input prices, technology, consumer income, population, tastes, and prices of related goods.

Shocks to exogenous variables shift supply and/or demand curves. The market then finds a new equilibrium where supply equals demand.

Formalizing the Model

Demand Curve

Step 1. Start with the elasticity definition

For demand, the own-price elasticity is defined as:

$$E_D = \frac{\% \Delta Q_D}{\% \Delta P}$$

This formula says: “a 1% change in price leads to E_D percent change in quantity demanded.” Remember that E_D is typically **negative** because price and quantity demanded move in opposite directions.

Step 2. Rearrange to express quantity changes

Multiply both sides by $\% \Delta P$ to solve for the percent change in quantity demanded:

$$\% \Delta Q_D = E_D \cdot (\% \Delta P)$$

This equation tells us how much quantity demanded changes in response to a given price change, holding everything else constant.

Step 3. Add in demand shocks

Markets can also shift for reasons other than price (income, tastes, population, etc.). We capture these **exogenous demand shifts** with a new term, S_D :

$$\% \Delta Q_D = E_D (\% \Delta P) + S_D$$

- $\% \Delta Q_D$: percent change in quantity demanded
- $\% \Delta P$: percent change in price
- E_D : own-price elasticity of demand (negative)
- S_D : demand shock (e.g., if incomes rise and demand increases by 7%, then $S_D = 7$)

Takeaway: This step-by-step process shows how we move from the basic elasticity definition to an equation that includes both **price effects** and **shifts in demand**.

Supply Curve

Step 1. Start with the elasticity definition

For supply, the own-price elasticity is defined as:

$$E_S = \frac{\% \Delta Q_S}{\% \Delta P}$$

This formula says: “a 1% change in price leads to E_S percent change in quantity supplied.” Unlike demand, E_S is typically **positive** because price and quantity supplied move in the same direction.

Step 2. Rearrange to express quantity changes

Multiply both sides by $\% \Delta P$ to solve for the percent change in quantity supplied:

$$\% \Delta Q_S = E_S \cdot (\% \Delta P)$$

This equation captures how much quantity supplied changes in response to a price change, holding everything else constant.

Step 3. Add in supply shocks

Supply can also shift due to exogenous factors (e.g., technology, input costs, regulations). We capture these **exogenous supply shifts** with a new term, S_S :

$$\% \Delta Q_S = E_S (\% \Delta P) + S_S$$

- $\% \Delta Q_S$: percent change in quantity supplied
- $\% \Delta P$: percent change in price
- E_S : own-price elasticity of supply (positive)
- S_S : supply shock (e.g., if new technology lowers costs and supply increases 5%, then $S_S = 5$)

Solving for the Equilibrium Price

Now we combine supply and demand.

1. In equilibrium, quantity supplied equals quantity demanded:

$$\% \Delta Q_S = \% \Delta Q_D$$

2. Substitute in the supply and demand equations:

$$E_S(\% \Delta P) + S_S = E_D(\% \Delta P) + S_D$$

3. Rearrange terms:

$$(E_S - E_D)(\% \Delta P) = S_D - S_S$$

4. Solve for the endogenous variable, price:

$$\% \Delta P = \frac{S_D - S_S}{E_S - E_D}$$

Intuition

- The denominator $(E_S - E_D)$ is positive since $E_S > 0$ and $E_D < 0$.
- If $S_D > 0$ (a positive demand shock, e.g., higher incomes) and $S_S = 0$, then $\% \Delta P > 0 \rightarrow$ price rises.
- If $S_S > 0$ (a positive supply shock, e.g., lower production costs) and $S_D = 0$, then $\% \Delta P < 0 \rightarrow$ price falls.

Takeaway: With both curves expressed in elasticity form, we can solve for how price responds to shocks. Once $\% \Delta P$ is known, we substitute back into either curve to find the equilibrium change in quantity.

Solving for the Equilibrium Quantity

Once we know the equilibrium price change $\% \Delta P$, we can solve for the change in quantity.

1. Substitute $\% \Delta P$ into the supply curve

Recall the supply curve:

$$\% \Delta Q_S = E_S(\% \Delta P) + S_S$$

Plug in the solved value for $\% \Delta P$:

$$\% \Delta Q_S = E_S \left(\frac{S_D - S_S}{E_S - E_D} \right) + S_S$$

2. Or substitute into the demand curve

Recall the demand curve:

$$\% \Delta Q_D = E_D(\% \Delta P) + S_D$$

Substitute the same $\% \Delta P$:

$$\% \Delta Q_D = E_D \left(\frac{S_D - S_S}{E_S - E_D} \right) + S_D$$

3. Confirm consistency

In equilibrium, both give the same answer:

$$\% \Delta Q_S = \% \Delta Q_D = \% \Delta Q$$

If they don't match, a calculation error has been made.

Intuition

- A positive demand shock ($S_D > 0$) raises price and increases equilibrium quantity.
- A positive supply shock ($S_S > 0$) lowers price but still increases equilibrium quantity.
- The exact size of the change depends on how responsive supply and demand are (their elasticities).

Takeaway: First solve for $\% \Delta P$ using equilibrium, then plug it into either curve to get $\% \Delta Q$. Both equations must agree at the new equilibrium.

Quick Recap

Once $\% \Delta P$ is found, substitute back into either curve:

- Supply:

$$\% \Delta Q_S = E_S(\% \Delta P) + S_S$$

- Demand:

$$\% \Delta Q_D = E_D(\% \Delta P) + S_D$$

Both give the same $\% \Delta Q$ in equilibrium.

General Steps for an EDM

1. **Identify shocks:** Determine S_S and S_D . At least one must be nonzero.

2. **Specify supply curve:**

$$\% \Delta Q_S = E_S(\% \Delta P) + S_S$$

3. **Specify demand curve:**

$$\% \Delta Q_D = E_D(\% \Delta P) + S_D$$

4. **Solve for price:** Set $\% \Delta Q_S = \% \Delta Q_D$ and solve for $\% \Delta P$.

5. **Solve for quantity:** Substitute $\% \Delta P$ into either equation to calculate $\% \Delta Q$.

Example: Impacts of Manure Regulations in the Pork Market

Pork producers are facing tougher regulations on how they manage and treat hog manure. These regulations raise production costs, which shifts the supply curve **leftward**, raising price and lowering quantity. This movement from one equilibrium to another is referred to as an **equilibrium displacement**.

Step 1. Represent the Shock

Suppose the current hog price is \$50/cwt, and the new regulations increase production costs by \$1/cwt.

- \$1 is **2%** of \$50.
- From the producer's perspective, paying \$1 more per cwt is like receiving \$1 less in price.

Thus, this regulation is equivalent to a **2% reduction in effective price**.

With supply elasticity $E_S = 2.15$:

$$S_S = E_S(-2) = 2.15(-2) = -4.3\%.$$

So the exogenous supply shock is a -4.3% shift in supply.

There is no demand shift: $S_D = 0$.

Step 2. Write Supply and Demand Equations

Supply:

$$\% \Delta Q_S = E_S(\% \Delta P) + S_S = 2.15(\% \Delta P) - 4.3$$

Demand:

$$\% \Delta Q_D = E_D(\% \Delta P) + S_D = -1.96(\% \Delta P) + 0$$

Step 3. Impose Equilibrium

In equilibrium, $\% \Delta Q_S = \% \Delta Q_D$:

$$2.15(\% \Delta P) - 4.3 = -1.96(\% \Delta P)$$

Rearrange:

$$(2.15 + 1.96)(\% \Delta P) = 4.3$$

Solve for price:

$$\% \Delta P = \frac{4.3}{2.15 + 1.96} = 1.05\%.$$

Step 4. Solve for Quantity

Plug back into supply or demand:

- Supply:

$$\% \Delta Q_S = 2.15(1.05) - 4.3 = -2.05\%$$

- Demand:

$$\% \Delta Q_D = -1.96(1.05) = -2.05\%$$

Result

- Price increases by 1.05%
- Quantity decreases by 2.05%

i Interpretation

Although regulations raised production costs by 2%, the market price rose by only about 1%. The remainder of the adjustment occurred through reduced output: pork production fell by about 2%.

This reflects how elasticities distribute the burden of shocks between **price changes** and **quantity changes**.

Step 5. Short vs. Long Run

This example uses **long-run elasticities** ($E_S = 2.15$, $E_D = -1.96$).

To examine the short-run impacts, simply replace these with short-run elasticities.

Pork Market: Supply Shift from Manure Regulations

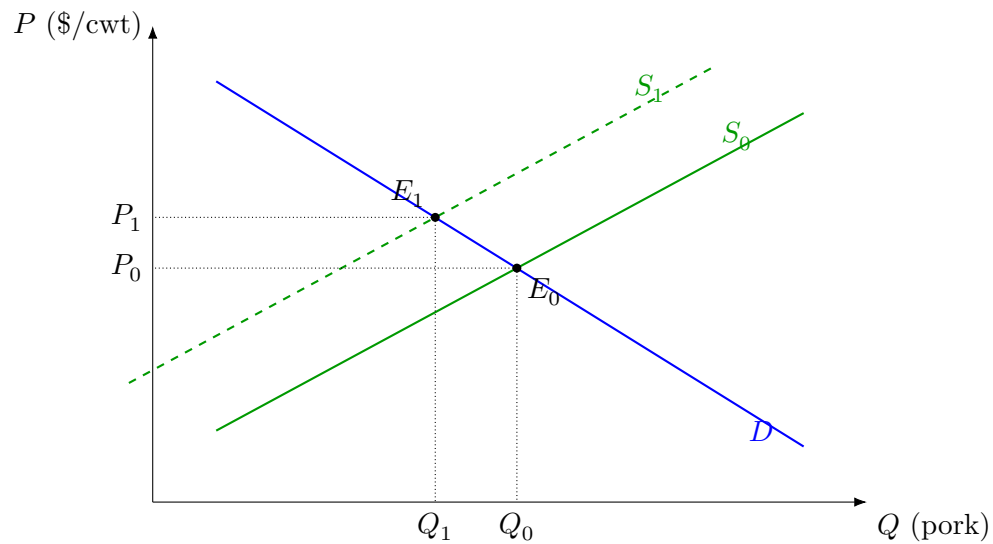


Figure 1: Manure regulations raise costs $\Rightarrow$ supply shifts left ($S_0 \rightarrow S_1$): P rises, Q falls.